

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2020

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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- 1. Let G be a finite group, and for each prime p dividing the order of G , let G_p be a fixed Sylow p -subgroup of G .
 - (a) Let H be the subgroup of G generated by the subgroups G_p , where p runs through the primes dividing the order of G . Show that $H = G$.
 - (b) Show that the following are equivalent:
 - I. Each subgroup G_p is normal in G .
 - II. For all $x, y \in G$, if x and y have relatively prime orders, then $xy = yx$.
- 2. Let M be the 3×3 complex matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 10 \\ 0 & 0 & 7 \end{pmatrix}.$$

- (a) Find the Jordan canonical form of M (you do not need to find the corresponding basis).
 - (b) The set of 3×3 complex matrices N such that $MN = NM$ forms a vector space V over $\mathbb{C}$. Find the dimension of V .
- 3. Let $\mathbb{Z}[\frac{1}{2}]$ be the abelian group consisting of rational numbers with only powers of 2 in their denominators, and let $\mathbb{Z}_{(2)}$ be the abelian group consisting of rational numbers with odd denominators.
 - (a) Show that every element of $\mathbb{Z}[\frac{1}{2}] \otimes_{\mathbb{Z}} \mathbb{Z}_{(2)}$ can be written in the form $a \otimes b$.
 - (b) Show that $\mathbb{Z}[\frac{1}{2}] \otimes_{\mathbb{Z}} \mathbb{Z}_{(2)} \simeq \mathbb{Q}$ as abelian groups. (*Note:* Don't forget to show that the map giving the isomorphism is well-defined.)

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4. Let R be a commutative ring with 1 and let $I_1, I_2, \dots, I_n$ be ideals of R such that $I_1 \cap I_2 \cap \dots \cap I_n = 0$. Prove that if each R/I_j is a Noetherian ring, then R is Noetherian.
5. Let $K/\mathbb{Q}$ be a Galois extension of degree 60 with $\text{Gal}(K/\mathbb{Q}) = A_5$, the group of even permutations of 5 letters.
- (a) Let $f(x) \in \mathbb{Q}[x]$ be a cubic polynomial that has a root in K . Show that $f(x)$ has a root in $\mathbb{Q}$.
- (b) Show that there is an irreducible polynomial $g(x) \in \mathbb{Q}[x]$ of degree 12 that has a root in K .
6. The group S_4 of permutations of 4 objects has 5 conjugacy classes: The identity, the six 2-cycles, the eight 3-cycles, the three products of two disjoint 2-cycles, and the six 4-cycles. There is a 4-dimensional representation $\rho_1 : S_4 \rightarrow GL_4(\mathbb{C})$ of S_4 such that

$$\begin{aligned} \rho_1((1, 2)) &= \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, & \rho_1((1, 2, 3)) &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ \rho_1((1, 2)(3, 4)) &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \rho_1((1, 2, 3, 4)) &= \begin{pmatrix} 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \end{aligned}$$

There is a 2-dimensional representation $\rho_2 : S_4 \rightarrow GL_2(\mathbb{C})$ of S_4 such that

$$\begin{aligned} \rho_2((1, 2)) &= \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, & \rho_2((1, 2, 3)) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \rho_2((1, 2)(3, 4)) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \rho_2((1, 2, 3, 4)) &= \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}. \end{aligned}$$

Show that each of ρ_1 and ρ_2 is the sum of two irreducible representations and there exist one-dimensional subrepresentations σ_j of ρ_j for $j = 1, 2$ such that $\sigma_1 \simeq \sigma_2$.